



KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI
POLITEKNIK NEGERI MALANG
JURUSAN TEKNOLOGI INFORMASI
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Mata Kuliah : Pemrograman Mobile
Program Studi : D4 – Teknik Informatika
Semester : 5

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Pertemuan : 2

LAPORAN PRAKTIKUM

Praktikum: layout sederhana (warm-up)

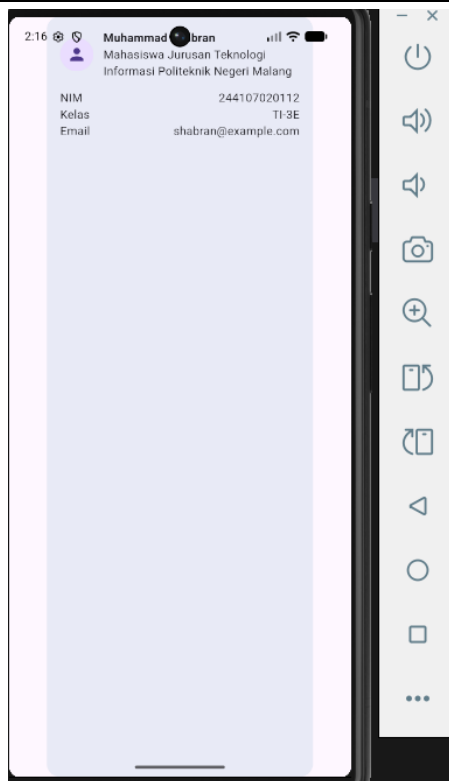
Langkah	Keterangan
1.	Membuat Layout Warm-up



	<pre>import 'package:flutter/material.dart'; void main() => runApp(const ProfileApp()); class ProfileApp extends StatelessWidget { const ProfileApp({super.key}); @override Widget build(BuildContext context) { return const MaterialApp(debugShowCheckedModeBanner: false, home: Scaffold(body: Center(child: ProfileCard()),),); } } class ProfileCard extends StatelessWidget { const ProfileCard({super.key}); @override Widget build(BuildContext context) { return Container(width: 320, padding: const EdgeInsets.all(16), decoration: BoxDecoration(color: Colors.indigo.shade50, borderRadius: BorderRadius.circular(16),), child: Column(mainAxisAlignment: MainAxisAlignment.min, children: [Row(children: [const CircleAvatar(child: Icon(Icons.person)), const SizedBox(width: 12),],), Row(children: [const CircleAvatar(child: Icon(Icons.person)), const SizedBox(width: 12), Expanded(child: Column(crossAxisAlignment: CrossAxisAlignment.start, children: const [Text('Muhammad Shabran', style: TextStyle(fontWeight: FontWeight.bold),), Text('Mahasiswa Jurusan Teknologi Informasi Politeknik Negeri Malang'),],),),],), const SizedBox(height: 12), const Row(children: [Expanded(child: Text('NIM')), Text('244107020112'),],), const Row(children: [Expanded(child: Text('Kelas')), Text('TI-3E'),],),],),); } }</pre>
2.	Cek Tampilan (
3.	Experimen 1 Menghapus Expanded pada Nama

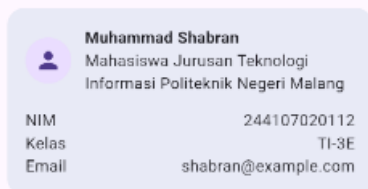


	<pre>Column(crossAxisAlignment: CrossAxisAlignment.start, children: const [Text('Muhammad Shabran', style: TextStyle(fontWeight: FontWeight.bold),), Text('Mahasiswa JTI Polinema'),],)</pre>  Menghapus widget Expanded pada elemen di dalam Row menghilangkan batasan maksimal ruang fleksibelnya. Jika panjang teks melebihi sisa ruang layar, teks tidak akan otomatis turun ke baris baru melainkan menembus batas dan menyebabkan <i>RenderFlex Overflow</i>. Expanded sangat penting untuk memaksa elemen beradaptasi hanya pada sisa ruang kosong yang tersedia secara aman.
4.	Experimen 2 :Ganti <code>mainAxisSize: MainAxisSize.min</code> menjadi nilai default dan amati perubahan tinggi kartu. <pre>child: Column(mainAxisSize: MainAxisSize.max,</pre>



Mengubah mainAxisSize: MainAxisSize.min menjadi nilai default-nya (max) membuat perilaku Column berubah dari yang awalnya hanya mengambil ruang vertikal sekecil mungkin (sebatas tinggi konten), menjadi merentang maksimal untuk mengisi seluruh sisa ruang vertikal layar. Akibatnya, elemen latar belakang kartu profil Anda kehilangan sifat membungkus kontennya secara rapat dan ikut meregang secara tidak proporsional menutupi ruang kosong di bawahnya.

5. Eksperimen 3 : Tambahkan satu baris data (misal Email) menggunakan pola Row + Expanded yang sama.





	<pre>const Row(children: [Expanded(child: Text('Kelas')), Text('TI-3E'),],) const Row(children: [Expanded(child: Text('Email')), Text('shabran@example.com'),],)</pre>
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Praktikum: dashboard responsif

Langkah	Keterangan
1.	Menyiapkan project <pre>active-ui-responsive-design> flutter create responsive_dashboard Creating project responsive_dashboard... Resolving dependencies in `responsive_dashboard`... (1.3s) Downloading packages... Got dependencies in `responsive_dashboard`. Wrote 131 files. All done! You can find general documentation for Flutter at: https://docs.flutter.dev/ Detailed API documentation is available at: https://api.flutter.dev/ If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev In order to run your application, type: \$ cd responsive_dashboard \$ flutter run Your application code is in responsive_dashboard\lib\main.dart.</pre> <pre>active-ui-responsive-design> cd responsive_dashboard PS C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course\02-week-2-declarative-ui-responsive-design\responsive_dashboard> flutter run Launching lib\main.dart on sdk gphone16k x86 64 in debug mode... WARNING: A restricted method in java.lang.System has been called WARNING: java.lang.System::load has been called by net.rubygrapefruit.platform.internal.NativeLibraryLoader in an unnamed module (file:/C:/Users/muham/.gradle/wrapper/dists/gradle-9.3.1-all/9ot9r568e8zfvd4mn8rbuij0/gradle-9.3.1/lib/native-platform-0.22-milestone-29.jar) WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module WARNING: Restricted methods will be blocked in a future release unless native access is enabled Running Gradle task 'assembleDebug'... 59.2s ✓ Built build/app/outputs/flutter-apk/app-debug.apk Installing build/app/outputs/flutter-apk/app-debug.apk... 13.7s I/FlutterActivityAndFragmentDelegate(5045): If you are attempting to set --enable-dart-profiling via Intent extras to launch a Flutter component outside of using the Flutter CLI, note that support for setting engine flags on Android via Intent will soon be dropped; see https://github.com/flutter/flutter/issues/180686 for more information on this breaking change. To migrate, set --enable-dart-profiling or any other flags specified via Intent extras on the command line instead or see https://github.com/flutter/flutter/blob/main/docs/engine/Flutter-Android-Engine-Flags.md for alternative methods. D/FlutterJNI(5045): Beginning load of flutter... D/FlutterJNI(5045): flutter (null) was loaded normally! I/flutter (5045): [IMPORTANT:flutter/shell/platform/android/android_context_gl_impeller.cc(104)] Using the Impeller rendering backend (OpenGL ES)</pre>



2. Memasukkan Kode Dashboard & Interaksi

```
import 'package:flutter/material.dart';
import 'package:flutter/cupertino.dart'; // Wajib ditambahkan untuk CupertinoSwitch

void main() => runApp(const DashboardApp());

class DashboardApp extends StatefulWidget {
  const DashboardApp({super.key});

  @override
  State<DashboardApp> createState() => _DashboardAppState();
}

class _DashboardAppState extends State<DashboardApp> {
  bool isDark = false; // Variabel penyimpan status tema

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        useMaterial3: true,
        colorSchemeSeed: Colors.indigo,
      ),
      darkTheme: ThemeData(
        useMaterial3: true,
        brightness: Brightness.dark,
        colorSchemeSeed: Colors.indigo,
      ),
      // Menerapkan tema berdasarkan nilai isDark
      themeMode: isDark ? ThemeMode.dark : ThemeMode.light,
      home: DashboardPage(
        isDark: isDark,
        // Fungsi untuk mengubah status tema ketika switch ditekan
        onDarkChanged: (value) => setState(() => isDark = value),
      ),
    );
  }
}
```

```
class DashboardPage extends StatelessWidget {
  const DashboardPage({
    required this.isDark,
    required this.onDarkChanged,
    super.key,
  });

  final bool isDark;
  final ValueChanged<bool> onDarkChanged;

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Student Dashboard'),
        actions: [
          Row(
            children: [
              Icon(isDark ? Icons.dark_mode : Icons.light_mode),
              const SizedBox(width: 4),
              // Komponen switch bergaya iOS (Cupertino)
              CupertinoSwitch(
                value: isDark,
                onChanged: onDarkChanged,
              ),
              const SizedBox(width: 12),
            ],
          ),
        ],
      ),
      // LayoutBuilder untuk responsivitas (Grid)
      body: LayoutBuilder(
        builder: (context, constraints) {
          final columns = constraints.maxWidth >= 700 ? 2 : 1;
        },
      ),
    );
  }
}
```

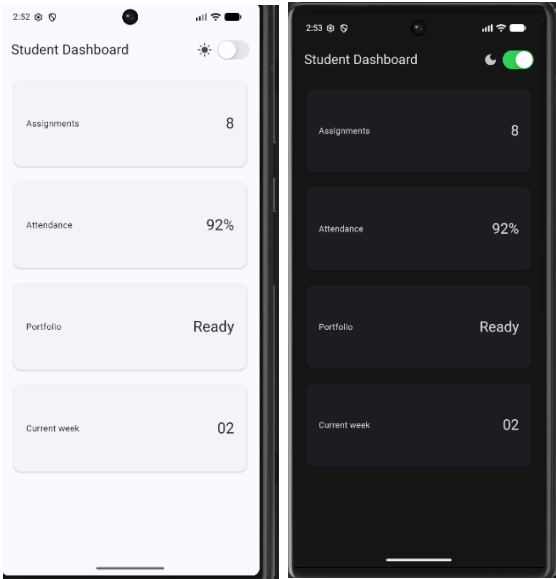
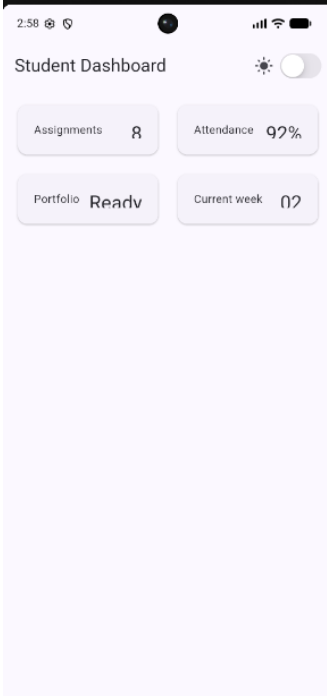


```
65         ],
66     ],
67 },
68 ],
69 },
70 // LayoutBuilder untuk responsivitas (Grid)
71 body: LayoutBuilder(
72   builder: (context, constraints) {
73     final columns = constraints.maxWidth >= 700 ? 2 : 1;
74     return GridView.count(
75       padding: const EdgeInsets.all(16),
76       crossAxisCount: columns,
77       crossAxisSpacing: 16,
78       mainAxisSpacing: 16,
79       childAspectRatio: 2.6,
80       children: const [
81         DashboardCard(title: 'Assignments', value: '8'),
82         DashboardCard(title: 'Attendance', value: '92%'),
83         DashboardCard(title: 'Portfolio', value: 'Ready'),
84         DashboardCard(title: 'Current week', value: '02'),
85       ],
86     );
87   },
88 ),
89 );
90 }
91 }
92
93 class DashboardCard extends StatelessWidget {
94   const DashboardCard({required this.title, required this.value, super.key});
95   final String title;
96   final String value;
```

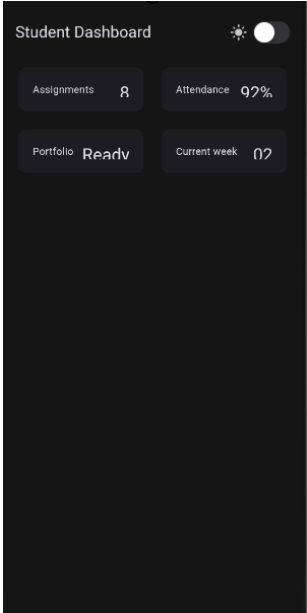
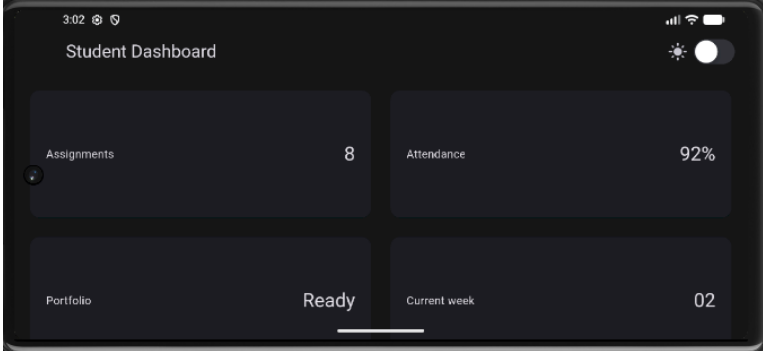
```
97   },
98   ),
99 );
100 }
101 }
102
103 class DashboardCard extends StatelessWidget {
104   const DashboardCard({required this.title, required this.value, super.key});
105   final String title;
106   final String value;
107
108   @override
109   Widget build(BuildContext context) {
110     return Card(
111       child: Padding(
112         padding: const EdgeInsets.all(20),
113         child: Row(
114           children: [
115             Expanded(child: Text(title)),
116             Text(
117               value,
118               style: Theme.of(context).textTheme.headlineSmall,
119             ),
120           ],
121         ),
122       ),
123     );
124   }
125 }
```

Hasil:



	
3.	Experiment 1 : Ubah breakpoint dari 700 menjadi nilai lain dan amati perubahan jumlah kolom. <pre>builder: (context, constraints) { final columns = constraints.maxWidth >= 300 ? 2 : 1; return GridView.count(</pre> 
4.	Experiment 2 : Ubah themeMode menjadi ThemeMode.dark, lalu kembalikan ke ThemeMode.system.



	<pre>themeMode: ThemeMode.dark, home: DashboardPage(Student Dashboard Assignments 8 Attendance 92% Portfolio Ready Current week 02</pre> 
5.	Experiment 3 : Uji aplikasi dengan ukuran layar emulator yang berbeda. 
6.	Experiment 4 : Tambahkan Semantics atau label yang bermakna pada elemen yang penting bagi screen reader. <pre>// Komponen switch bergaya iOS yang dibungkus Semantics → Tab to Jump Semantics(label: isDark ? 'Ubah ke mode terang' : 'Ubah ke mode gelap', child: CupertinoSwitch(value: isDark, onChanged: onDarkChanged,),),</pre>

